

Mohammad Nakhaei Ramezanzadeh

AI Engineer & Data Scientist

PyTorch · Computer vision · Test-time adaptation · Statistics · LLMs

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PROFILE

AI engineer and researcher with a physics foundation from Sharif University of Technology. Author of ongoing research that turns **cybernetic feedback control** into a method for **safe test-time adaptation**, so far beating eight baselines on continual CIFAR-100-C and ImageNet-C. Writes reproducible **PyTorch** experiments, builds statistically honest backtests, integrates LLMs, and ships it all in production web apps (**17 live products** since 2023).

RESEARCH

The Cybernetic Intelligence Functional: Predictive-Diversity Collapse as a Unified Monitor and Controller for Safe Test-Time Adaptation

Sole author · in preparation, experiments ongoing · 2026

- Wraps test-time adaptation in a feedback controller on **predictive diversity**, with an Ashby-style freeze/reset to stable BN-adapt when diversity collapses.
- Results so far: no collapse on small-batch continual streams, with **41.81 ± 0.07%** error on CIFAR-100-C (5 seeds) vs 44.18% for the best baseline, and **60.30%** vs 68.42% (SAR) on ImageNet-C.
- Re-implemented 8 baselines (TENT, CoTTA, EATA, SAR, MEMO, LAME and more) in one reproducible PyTorch codebase, with multi-seed Kaggle GPU runs, ablations and negative results.

The Requisite-Variety Law: A Cybernetic Principle of Self-Monitoring Against Collapse in Learning Systems

Sole author · in preparation · one law of self-monitoring, being tested across five domains

EXPERIENCE

Full-Stack & AI Developer · **Freelance** Remote, Tehran — Jul 2023 – Present

- Integrated AI and machine-learning features into client web products with **PyTorch**, **TensorFlow** and **OpenCV**, and launched 17 products end to end on Next.js, NestJS and PostgreSQL.

Backend Developer · **SAV** Tehran · finance & investment — Feb 2025 – Jun 2025

- Built and maintained **Laravel** REST APIs and tuned **MySQL** performance for a finance and investment product.

PROJECTS

Prophet — private repository

Quant research and trading engine: backtests on Binance data since 2020 and FX and gold since 2003, with bootstrap CIs, deflated Sharpe and walk-forward tests.

Rust · statistics · time series

GLM Toolkit — personal project

Typed SDK, CLI, REST server and web UI for the GLM model family: text reasoning, vision and image generation, with streaming.

TypeScript · LLM APIs

PDF-to-Text — github.com/MNakhaeiR ↗

Text extraction with automatic fallback from PyPDF2 to pdfplumber to Tesseract OCR for scanned documents.

Python · OCR

Professor Analysis — t.me/OstadshenasiTJ_bot ↗

Telegram bot that helps university students look up and evaluate professors before choosing courses.

Python · Telegram Bot API

SKILLS

ML & DL	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV
Methods	Computer vision, distribution shift and test-time adaptation, NLP, deep learning, persistent homology
Data & stats	Python, NumPy, Pandas, Matplotlib, SQL, ETL, bootstrap and multi-seed statistics, A/B testing
LLMs	API integration (text, vision, image generation, streaming), local inference with Ollama
Deployment	Docker, FastAPI, Flask, REST APIs, GitHub Actions, Kaggle GPUs, Linux; also TypeScript, Rust, LaTeX

EDUCATION

B.Sc. Computer Engineering — Sep 2024 – Present

Islamic Azad University, South Tehran Branch

Physics (undergraduate) — Sep 2020 – Jun 2023

Sharif University of Technology · left to pursue computer engineering

LANGUAGES

Persian	Native
English	Professional
German	Basic